SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA15

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.* (BL4)

Dave's Taxonomy: Articulation (Level 4) **Total Marks: 40**

Question Count: 5

Duration :60 Minutes

Code of Conduct

I _G.Althaf(192372004) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student G.Althaf(192372004)

Assessment Tasks

Industry Problem Solving Task

1. **Retail Data Optimization Problem -** A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.

2. **Social Media Fraud Detection Problem -** A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.

3. **Government Data Platform Problem** A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.

4. **Banking Fraud Detection Problem**
A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data.
Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement.

5. **Agriculture Smart Farming Problem**
   A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

Industry Problem Solving Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Understanding of Industry Context	20%	Demonstrates strong understanding of real-world problem context	Good understanding	Basic understanding	Weak understanding
Application of Big Data Concepts	25%	Applies highly relevant big data ideas and tools	Mostly appropriate application	Partial application	Weak application
Solution Design	25%	Solution is practical, scalable, and well-reasoned	Reasonable solution	Basic solution with gaps	Unclear solution
Security / Risk Awareness	15%	Strong awareness of security and operational risks	Reasonable awareness	Limited awareness	Poor awareness
Clarity and Professionalism	15%	Well-structured and professional response	Generally clear	Moderately clear	Poorly presented

Total Marks Awarded:

<u>Problem 1: Retail Data Optimization System</u>

1. Architectural Design & Ingestion Pipeline

- **Ingestion Layer:** Point-of-Sale (POS) store terminals, e-commerce web/mobile clickstreams, and IoT RFID shelf scanners stream real-time events to **Apache Kafka**. Batch extracts from supplier ERPs and legacy inventory systems are ingested periodically using **Apache NiFi** or **Airflow**.

- **Storage Layer:** A decoupled Data Lakehouse architecture utilizing **Delta Lake** on top of **Amazon S3** or **Azure Data Lake Storage (ADLS)**. Data is segmented into Bronze (raw JSON clickstreams and POS logs), Silver (cleansed, deduplicated, and unified sales/inventory tables), and Gold (aggregated customer profiles and regional inventory metrics).

- **Serving & Caching Layer: Redis Enterprise** or **Apache Cassandra** caches real-time inventory counts and customer interaction vectors to serve low-latency queries during customer checkout and browsing.

2. Tools, Technologies & Processing Engines

- **Stream Processing Engine: Apache Flink** calculates real-time sliding-window inventory balances across localized fulfillment centers and triggers instant restock notifications.
- **Distributed Batch Processing: Apache Spark (Core & SQL)** executes nightly data merges, customer lifetime value (CLV) calculations, and complex basket analysis.
- **Distributed Query Engine: Trino (Presto)** or **ClickHouse** enables multi-region interactive BI reporting on top of the Lakehouse.

3. Analytics & Machine Learning Models

- **Demand Forecasting:** Hybrid time-series models combining **Prophet** and **DeepAR / Temporal Fusion Transformers (TFT)** analyze multi-seasonality, local promotions, and regional weather factors to predict SKU-level demand.
- **Personalized Recommendations:** Two-stage recommendation pipeline:
    - *Candidate Generation:* Approximate Nearest Neighbors (**FAISS**) over embeddings generated from collaborative filtering.
    - *Real-Time Ranking:* Gradient Boosted Decision Trees (**XGBoost / LightGBM**) combining live browsing context from a feature store (**Feast**) with historical purchases.

4. Security and Risk Mitigation

- **Payment Security:** Full compliance with **PCI-DSS**; tokenization of credit card records and cryptographic salting of personally identifiable information (PII) before storage.
- **Access Control:** Attribute-Based Access Control (**ABAC**) via **Apache Ranger** ensures regional store managers only access their local store inventory, while central marketing teams access aggregated, anonymized analytical datasets.

Problem 2: Social Media Fraud and Misinformation Detection

1. Scalable Big Data Architecture

- **High-Throughput Ingestion:** Millions of real-time posts, comments, likes, and graph interactions are ingested through multi-partition **Apache Kafka** clusters.
- **Stream Processing Backbone: Apache Flink** consumes high-velocity topics with sub-second latency, implementing stateful stream operations to compute interaction velocities (e.g., sudden bursts in post frequency or bot-like liking rates).
- **Graph Storage Tier: Neo4j** or **Amazon Neptune** maintains live follower/following connection graphs and co-engagement topologies to uncover coordinated bot networks and sybil clusters.

2. Machine Learning and Content Intelligence

- **Text & Media Misinformation Models:** Fine-tuned Transformer models (e.g., **RoBERTa / DeBERTa**) classify textual misinformation and sensationalist claims against verified fact-check knowledge bases. Multimodal Vision Transformers evaluate manipulated media/deepfakes.
- **Bot & Fake Account Detection:** Graph Neural Networks (**GNNs**, such as Graph Convolutional Networks) examine graph structural anomalies (e.g., dense subgraphs with high out-degree ratios), combined with **Isolation Forests** for anomalous temporal behavioral patterns.

3. Real-Time Decisioning Pipeline

- **Feature Serving:** Feature store (**Feast**) serves online user reputation scores to inference engines with < 15 ms latency.
- **Multi-Tier Filtering:**
    - *L1 (Edge Filter):* Deterministic rule engines block known malicious URLs, spam phrases, and extreme rate-limit violations instantly.
    - *L2 (Streaming ML):* Flink-integrated **Triton Inference Servers** run lightweight embeddings for suspicious account flagging.
    - *L3 (Asynchronous Deep Scan):* High-compute deep neural networks analyze high-risk content in parallel queues before indexing to the public feed.

4. Fault Tolerance and Operational Risk

- **State Management:** Flink's asynchronous RocksDB state backend with persistent Chandy-Lamport checkpointing guarantees exactly-once processing across broker failures.
- **Moderation Audits:** Questionable or borderline classifications are routed to human-in-the-loop (HITL) review queues, preserving transparency and mitigating false-positive penalties against legitimate users.

Problem 3: Federated National Government Data Platform

1. Data Architecture & Multi-Agency Interoperability

- **Data Fabric / Mesh Architecture:** A domain-driven distributed data mesh where individual ministries (Census, Finance, Geospatial/Survey) manage their operational data domains while exposing federated data products.
- **Standardization & Metadata Management: Apache Atlas** for enterprise metadata catalogs, enterprise data lineage, and semantic definitions. Standardized data schemas (**DCAT-AP**, **GeoJSON**, **Parquet**) guarantee semantic interoperability across disparate agency systems.
- **Federated Query Engine: Trino** or **Apache StarRocks** executes cross-departmental federated queries across distributed agency databases without requiring physical data centralization.

2. Data Governance Strategy

- **Master Data Management (MDM):** Centralized identity resolution links citizen and enterprise records using cryptographic deterministic identifiers without storing raw national IDs in plain text.
- **Data Lifecycle & Retention:** Automated policy engines govern data lifecycle states (active, archival, destruction) based on legal national archival frameworks.

3. Security, Privacy, and Confidentiality

- **Privacy-Preserving Computation:** Differential privacy (ϵ, δ-differential privacy) is injected into public census and economic query engines to prevent individual re-identification attacks during statistical aggregation.
- **Encryption Standards: AES-256-GCM** encryption for all stored records, with hardware security modules (HSM) managing cryptographic keys, and mandatory **mTLS 1.3** for all intra-agency data pipelines.
- **Granular Access Control:** Role-Based and Attribute-Based Access Control (**RBAC / ABAC**) via **Apache Ranger**, enforcing dynamic column-masking (e.g., masking income and names) and row-level filtering based on user clearance level.

4. Auditability and Compliance

- **Immutable Audit Logging:** Every read, query, and export is recorded in append-only distributed ledger platforms or immutable WORM (Write Once, Read Many) cloud object storage, ensuring regulatory oversight and compliance against data leakage.

Problem 4: Ultra-Low-Latency Banking Fraud Detection Platform

1. High-Performance Streaming Pipeline

- Data Ingestion: Banking transaction streams (ATM, SWIFT, POS, online payment gateways) stream into Apache Kafka deployed across active-active redundant clusters.
- Real-Time Stream Processing: Apache Flink performs stateful sliding-window aggregations (5-second, 1-minute, and 24-hour windows) to compute rapid velocity metrics (e.g., transaction count per minute, geographic displacement speed).
- High-Speed State & Historical Store: Aerospike or Redis Enterprise provides sub-millisecond retrieval of historical customer behavioral baselines (typical spending locations, device footprints, merchant categories).

2. Hybrid Detection Algorithms & Analytics Models

- Rule-Based Pre-Filtering: Deterministic expert systems (Drools engine) evaluate strict regulatory checks, blacklist lookups, and velocity limit violations (< 5 ms).

- Supervised Machine Learning: LightGBM and XGBoost models trained on historical fraud patterns evaluate complex non-linear feature interactions to produce an instant fraud probability score.

- Unsupervised Anomaly Detection: Autoencoders and Isolation Forests flag zero-day fraud patterns and sudden structural deviations from a customer's historical baseline.

- Graph Pattern Analytics: Graph databases (Amazon Neptune / TigerGraph) detect mule account rings and coordinated merchant skimming networks.

3. Latency Optimization & Architecture Flow

$$\text{Incoming Transaction} \xrightarrow{<2\text{ ms}} \text{Kafka Ingestion} \xrightarrow{<10\text{ ms}} \text{Flink Feature Extraction} \xrightarrow{<15\text{ ms}} \text{ONNX/Triton ML Scoring} \xrightarrow{<5\text{ ms}} \text{De}$$

- Inference Serving: Models exported to optimized ONNX Runtime or TensorRT deployed on Triton Inference Server guarantee total end-to-end evaluation latency of under 40 ms, well within standard payment authorization windows (< 300 ms).

4. Risk Management and High Availability

- Fault Tolerance: Distributed active-active replication with automatic failover guarantees 99.999% system availability.

- Explainability & Auditing: Machine learning decisions generate local feature attribution values using TreeSHAP, providing immediate explainability codes for compliance teams and customer support during transaction blocking.

Problem 5: Big Data Architecture for Precision Smart Farming

1. Distributed Edge-to-Cloud Big Data Architecture

- Edge Processing: Farmland IoT gateways and drone ground stations run lightweight runtimes (K3s / EdgeX Foundry) to aggregate telemetry from soil moisture probes, ambient temperature sensors, and multispectral drone cameras locally, filtering out transmission noise before sending data uplink.

- Messaging Tier: Lightweight protocols (MQTT / CoAP) transmit edge readings to centralized EMQX brokers, bridging data directly into Apache Kafka.

- Hybrid Storage Tier:

    - *Time-Series Data:* TimescaleDB or InfluxDB stores high-frequency environmental telemetry (soil pH, moisture, humidity, solar radiation) indexed by timestamp and geo-sensor coordinates.

- *Spatial & Raster Storage:* Cloud Object Storage (Amazon S3 / MinIO) stores high-resolution multispectral drone imagery, historical satellite rasters (Sentinel-2 / Landsat), and elevation maps.

2. Geospatial Processing & Spatial Analytics

- Spatial Compute Engine: Apache Sedona (formerly GeoSpark) and PostGIS execute distributed spatial joins and overlay computations, mapping drone-derived vegetation index vectors against specific field boundaries and parcel ownership maps.
- Computer Vision Pipeline: Convolutional Neural Networks (U-Net / YOLOv8) process drone multispectral imagery to compute Normalized Difference Vegetation Index (NDVI) and Normalized Difference Water Index (NDWI), identifying early localized weed infestations and crop nutrient deficiencies.

3. Predictive Analytics for Decision Support

- Smart Irrigation Optimization: Reinforcement learning and evapotranspiration models combine live soil moisture metrics with multi-day predictive weather models to automate dynamic variable-rate irrigation controllers, cutting water usage while preventing crop stress.
- Yield Prediction & Harvest Planning: Gradient boosting algorithms (LightGBM) integrate historical yield data, current vegetative growth curves, and weather trajectories to provide farmers with harvest date recommendations and accurate volume projections.

4. Operational Resilience and Security

- Offline Resilience: Edge gateways feature local caching mechanisms (SQLite / local file queues) to ensure uninterrupted data collection during intermittent rural network connectivity.
- Data Integrity & Security: Sensor-to-cloud payload signing prevents malicious telemetry spoofing, while role-based encryption guarantees that commercial farm telemetry and yield metrics remain confidential to the respective landowners.